



TECH X ACADEMY

LEARN | GROW | INNOVATE

COURSE OUTLINE

Diploma in Machine Learning & Applied AI

Duration	6 Months (24 Weeks)
Contact Hours	72 Hours (3 Classes/Week × 1 Hour/Class)
Target Audience	Matric-pass students (Beginners)
Prerequisites	Basic computer literacy; no prior coding or advanced math required.



1. EXECUTIVE SUMMARY

This diploma is designed to take students from zero knowledge to job-ready Machine Learning (ML) practitioners. The curriculum prioritizes practical ML engineering — data handling, modeling, evaluation, and deployment — while integrating the essential mathematics (Linear Algebra, Calculus, Statistics) required to understand how algorithms work. The course includes brief introductory modules on Deep Learning, Computer Vision, and NLP to provide industry context, culminating in a capstone project and portfolio development.

2. LEARNING OBJECTIVES

By the end of this program, students will be able to:

- **Code in Python:** Write clean, functional Python scripts for data analysis and automation.
- **Apply Mathematics:** Utilize linear algebra, calculus, and statistics to interpret model behavior and optimize performance.
- **Build ML Pipelines:** Execute end-to-end workflows from data cleaning and feature engineering to model training and evaluation using scikit-learn.
- **Evaluate Models:** Critically assess model performance using appropriate metrics (Accuracy, Precision, Recall, F1, ROC-AUC) and avoid common pitfalls like overfitting and data leakage.
- **Deploy Solutions:** Package models and create simple web interfaces using Streamlit or Gradio.
- **Practice Ethics:** Identify and mitigate bias in data and models, adhering to responsible AI principles.

3. CURRICULUM STRUCTURE

Phase 0: Orientation & Setup (Week 0)

Goal: Establish the learning environment and baseline expectations.

Orientation

- Introduction to AI vs. ML vs. DL.
- Course roadmap and ethical guidelines.

Lab: Environment setup (Python, VS Code/Jupyter, Git) and "Hello World" notebook.

Phase 1: Foundations — Python & Math for ML (Weeks 1–4)

Goal: Build the coding and mathematical bedrock required for algorithms.

Week 1: Python Basics I

- Variables, data types, I/O, conditionals, loops.

Lab: Mini-calculator and number-guess game.

Week 2: Python Basics II

- Functions, lists, dictionaries, modules, file I/O (CSV/JSON).

Lab: Text adventure game and CSV statistics utility.

Week 3: Math for ML I — Linear Algebra

- Scalars, vectors, matrices, dot products, norms.

Lab: NumPy operations; implementing dot product and cosine similarity.



Week 4: Math for ML II — Calculus & Statistics

- Derivatives/gradients (optimization intuition), mean/variance, distributions, correlation.

Lab: Gradient descent on a simple line fit; exploring distributions.

Phase 2: Data Skills & Core ML (Weeks 5–10)

Goal: Master data manipulation and implement foundational ML algorithms.

Week 5: Data Manipulation

- NumPy arrays, Pandas DataFrames, filtering, grouping, merging.

Lab: Cleaning a raw dataset (e.g., student marks or sales data).

Week 6: Visualization & EDA

- Matplotlib/Seaborn, distributions, correlations, outliers.

Lab: Exploratory Data Analysis (EDA) notebook with 5 key insights.

Week 7: How ML Works

- Features/labels, train/test split, overfitting, KNN, Linear Regression.

Lab: First scikit-learn pipeline (e.g., Iris dataset).

Week 8: Classification & Metrics

- Logistic Regression, Decision Trees, Confusion Matrix, Precision/Recall/F1.

Lab: Predicting pass/fail or customer churn.

Week 9: Model Evaluation

- Cross-validation, ROC/AUC, bias-variance tradeoff.

Lab: Comparing 3 models using cross-validation.

Week 10: Feature Engineering

- Handling missing values, encoding categoricals, scaling, pipelines, data leakage.

Lab: Building a reusable preprocessing pipeline.

Phase 3: Applied ML & Projects (Weeks 11–16)

Goal: Apply advanced techniques and complete independent mini-projects.

Week 11: Regularization & Ensembles

- Ridge/Lasso, Random Forest, Gradient Boosting (XGBoost/LightGBM).

Lab: Improving baseline models with ensemble methods.

Week 12: Unsupervised Learning

- K-Means clustering, PCA (dimensionality reduction).

Lab: Customer segmentation or topic grouping.

Week 13: Mini-Project 1 (Supervised)

- Task: Define problem → EDA → Model → Evaluate → Report.

Week 14: Mini-Project 2 (Classification/Regression)

- Task: Focus on feature engineering and error analysis.

**Week 15: Ethics & Responsible AI**

- Bias in data/models, fairness metrics, privacy.

Lab: Auditing a model for bias; writing a responsible-use checklist.

Week 16: ML System Basics

- Saving/loading models (joblib/pickle), reproducibility (seeds, versioning).

Lab: Packaging a model and writing a prediction script.

Phase 4: Intro to Advanced Fields (Weeks 17–20)

Goal: Provide high-level exposure to Deep Learning, CV, and NLP.

Week 17: Neural Networks (Intro)

- Perceptrons, layers, activation functions, backpropagation intuition.

Lab: Training a small NN on tabular data.

Week 18: Computer Vision (Intro)

- Images as tensors, CNN intuition, transfer learning.

Lab: Image classifier demo using a pretrained model.

Week 19: NLP Fundamentals (Intro)

- Tokenization, TF-IDF, sentiment analysis.

Lab: Text classifier on reviews (scikit-learn).

Week 20: Deployment (Light Ops)

- Streamlit/Gradio for web demos, basic logging.

Lab: Deploying one mini-project as a web app.

Phase 5: Capstone & Career Prep (Weeks 21–24)

Goal: Synthesize learning into a portfolio-ready project.

Week 21: Capstone Scoping

- Problem definition, success metrics, data sourcing.

Week 22: Capstone Build I

- EDA, baseline model, feature engineering plan.

Week 23: Capstone Build II

- Final model tuning, evaluation, UI development.

Week 24: Demo Day

- Final presentations, GitHub repo polish, resume/LinkedIn basics.

4. ASSESSMENT STRATEGY

Component	Weight	Description
Quizzes	15%	Weekly concept checks to ensure theoretical understanding.



Component	Weight	Description
Mini-Projects	45%	Six to eight graded notebooks focusing on specific skills (e.g., EDA, Classification, Clustering).
Capstone Project	30%	A comprehensive end-to-end solution with code, report, and demo.
Participation & Ethics	10%	Engagement in labs and completion of responsible AI reflections.

5. TECHNOLOGY STACK

- **Language:** Python 3.x
- **Core Libraries:** NumPy, Pandas, Matplotlib, Seaborn, scikit-learn
- **Deep Learning (Intro):** TensorFlow/Keras or PyTorch
- **Deployment:** Streamlit or Gradio
- **Environment:** Jupyter Notebooks, Google Colab, VS Code, GitHub

6. INSTRUCTOR GUIDELINES

Class Structure

- Class 1: Concept + Live Demo (45 min) + Q&A (15 min).
- Class 2: Guided Lab (Code-along) with checkpoints.
- Class 3: Project work, code review, and short quiz.

Homework: Assign 2–3 micro-tasks (20–30 mins each) between classes to reinforce concepts.

Feedback: Provide specific feedback on code correctness, clarity, and next steps for improvement.

7. EXPECTED OUTCOMES

Upon completion, students will possess a GitHub portfolio containing 6–8 ML projects and 1 capstone, demonstrating proficiency in data analysis, model building, and deployment suitable for entry-level roles or further specialization.